

8

Logistic Regression and
the Generalized Linear Model

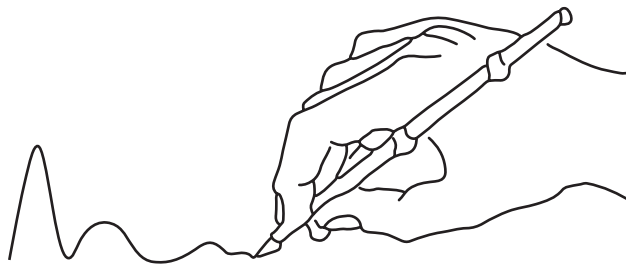
CHAPTER OBJECTIVES
■ Learn why dependent variables that are not continuous pose challenges for traditional ordinary least-squares regression analysis and why the generalized linear model is better equipped to deal with such variables. ■ Why logistic regression is well-suited to deal with binary response variables, either naturally-occurring or artificially created. ■ Understand the difference between the odds and the logarithm of the odds, known as the logit. ■ How to run a logistic regression in Python and interpret the most relevant output. ■ How to run a multiple logistic regression in Python and interpret odds for a variable in the context of other predictors.

Thus far in this book, we have considered a number of different models and statistical methods such as *t*-tests, ANOVA, regression, among others. While at first glance it may not be apparent, the fact of the matter is that most of the models we have considered thus far have assumed the response variable to be more or less **continuous** in nature. For example, when we modeled verbal scores as a function of quantitative scores, we assumed that the verbal variable was drawn from a population distribution that is more or less continuously distributed. That did not imply that verbal had to be normally distributed. It did not. However, it had to have **sufficient distribution in its scores** such that we could at least think of it as measurable along a continuous scale.

Recall the definition of continuity as defined by mathematicians discussed earlier in the book. Though calculus and real analysis define continuity quite formally and very rigorously, the essence of continuity is that any score for a variable is possible along the infinitely dense range of the variable. For example, for an IQ variable to be continuous in the range of, say, 0 to 200, it implies that any score within these limits is theoretically possible. That is, one could obtain a score of 95.02748 as a possible IQ, just as one could obtain a score of 110.08375, and so on for an infinite number of decimal places. And as discussed earlier in the book, true continuity in research variables

usually has to be **assumed** rather than the variable actually possessing a true continuous distribution. Why is this the case? Simply because **true continuity in its deepest theoretical sense implies that any particular value in such a distribution has a probability of zero of occurring**. This is the case since if you reach into a distribution deemed continuous, and if the variable is actually truly continuous, then it should be impossible to actually “catch” or “grab” any particular value. Akin to snorkeling in the weeds under water (imagine you are wading through the weeds as a deep-diver would), it becomes impossible to grab any of the weeds when you reach for a single one. The flow of the water has the effect of moving the particular weed aside just as you reach for it. In other words, it is impossible to catch or secure a particular value. Hence, when we dive into a distribution of a continuous mathematical variable, we likewise cannot secure a particular value. More formally, we say that **in the limit, the probability of any particular value is equal to 0**. As before, the idea of limits is beyond the scope of this book, but you can think of a limit in this case that as the “slices” toward the particular value get smaller and smaller, that is, as we “narrow in” on the distinct value, the probability of that value being obtainable **shrinks** toward zero (i.e. it has a limiting value of zero). In an analogous sense, as you walk toward a flat wall, the distance between you and the wall gets smaller and smaller, until in the limit, any distance greater than zero has a probability of zero. This is a powerful way to conceive of the limit concept.

An even more intuitive definition of continuity and one that everyone is familiar with is that of not lifting one’s pencil off the page as one draws a straight line from one endpoint to another. The line is deemed continuous if at no point along the line is the pencil lifted, as shown:



As discussed earlier in the book, whether one is working with continuous or non-continuous variables is a **major decision node** to deciding which analysis one should perform. For instance, if you ever work with a statistical or methodological consultant, you may spend a great deal of time in the planning phase of your experiment or study on the **measurement problem**, which is all about deciding how best to assess what you are seeking to measure. Measurement should not be taken for granted, and especially in sciences that are less “hard,” such as psychology, education, sociology, economics, and others, the measurement problem should require a great deal of your attention. Far too often students and researchers skip the measurement problem and proceed directly to performing statistical analyses on empirically measured variables that, when critically evaluated, likely do a poor job at assessing the underlying construct(s). This is not the complete fault of researchers, since many constructs in such sciences are inordinately difficult (or even impossible) to measure and often

require self-report to measure them, which is laden with difficulty since people are not always (seldom?) the best judge of their own behavior.

To carry on with our example, if you are to measure something like intelligence, how best should you measure it? Among the several issues you will encounter will be whether to treat the response (dependent variable) as either a continuous variable (at least on a theoretical level) or a non-continuous one. For instance, will you seek to measure individuals' precise scores on IQ or will you attempt to simply **classify** them into "low," "medium," and "high" IQ? How you **operationalize** the measurement of your variable will likely have a huge impact and influence on not only what statistical model you choose for your analysis, but also regarding what conclusions can be drawn or not drawn from the research. In fact, **one reason why some studies find effects and some do not may have less to do with whether a phenomenon truly exists in the population and more to do with how the variable was operationalized!** This itself should make understanding statistical modeling paramount to your work. Though a complete discussion of this sort is well beyond the scope of this book, you should never take the results of a study at "face value." Instead, always "dig" into how variables were **measured, defined, operationalized**, etc., as this could play a huge factor in determining whether the study yielded fruitful findings. How one study measures a variable may not even be remotely similar to how another study measures it. Hence, when you read the headline, "Those with higher IQ are more likely to earn more money," or some such, that statement or conclusion is only as good as the measurement that went into the study. **Be critical.** Even in medical and biological fields debates over treatment effectiveness from study to study can be quite difficult to discern as a result of the myriad of methodological factors influencing the given study. As we alluded to earlier in the book with regards to the pandemic of 2020, what does the number of deaths "due to COVID" actually mean? How are those dying **from** COVID identified versus those dying **with** COVID but ultimately due to a different underlying condition? How is the **causal mechanism** identified, if at all, regarding the cause of death? These questions are important! Complex statistical analyses on poorly conceived or measured variables is at best meaningless and at worst unethical and misleading. We explore this question a bit further in what follows.

8.1 How Are Variables Best Measured? Are There Ideal Scales on Which a Construct Should Be Targeted?

As mentioned, a full discussion of measurement is well beyond the scope of this book. For our purposes here, we wish only to comment a bit further and deeper on the measurement issue given that the model we will consider shortly will assume a binary-distributed response. However, before assuming our variable binary, the question naturally arises on how best to measure variables in general. As mentioned, this is a central problem in fields such as psychology and other social sciences where the measurement question is, or least should be, the most difficult but urgent methodological issue to resolve.

Should one attempt to measure variables on a continuous scale or should you "bin" them into categories? While there is no definitive and easy answer to this question, as a rough guideline, in most cases **measurement on a continuous scale is probably**

the best ideal to shoot for. Now, there is a good reason why “rough guideline” is mentioned here. In some cases, measurement on a continuous scale can do more harm than good and is definitely not appropriate, even if at first glance it may appear as an attractive option. In most cases, however, it is usually best to side with continuity or at least “pseudo-continuity” than implement a dichotomous or polytomous choice. Yet for some scales, such as **Likert scales** in which respondents are asked to rate their opinion or preference by selecting one of a range of whole numbers, one must be cautious about extracting too much meaning out of difference choices if the phenomena you are measuring has a great deal of measurement error or “random shocks” (i.e. chance) to it.

As an example, one that can be generalized to many Likert scales, suppose you would like to measure how much a person enjoys pizza. While you can informally ask people how much they enjoy it, you know that seeking scientific measurement of the phenomenon is a wiser approach. Hence, you decide to construct a **Likert scale** with values ranging from 0 (dislike pizza) to 10 (enjoy pizza immensely, eat it all day). Now, what do these numbers mean? In the complete abstract sense, there is no debate. That is, the real line with integers on it clearly designates a scatter of different numbers. However, you are not doing mathematics. You are doing science, and things get a whole lot more complicated (in this sense only) fast. For example, if a participant rates “0” while another rates “10,” you can probably be quite certain, assuming your participants are not lying, that one person enjoys pizza more than the other. It seems reasonable that the person circling 10 is a pizza lover, while the one circling 0 is likely not. We could probably be quite certain of this distinction.

However, if one person rates 6 while another 7, can you say the same thing? Can you even draw a similar conclusion but to a lesser degree? While numerically they have circled different choices, **whether one actually enjoys pizza more than the other should not be assumed simply because one rated higher than the other.** Doing so would constitute a major mistake and is one reason (among many) why social science is sometimes looked up with suspicion. It may be that these two individuals actually enjoy pizza to the same exact degree in reality, but may have differing **response styles**. By “response styles” here, we mean some people like to circle higher numbers than others to express their pizza-liking, even if they like pizza to the same degree (or even less) as the person circling a lesser score. Given this, how then could we ever know for certain who enjoys pizza more? Probably only through **behavioral observation**. That is, frequency of pizza eating over a given period of time or other behavioral measure would probably provide much more realistic and precise insight into true pizza preference than actually doing something quite trivial and asking subjects to respond to a questionnaire and assuming an abstract numerical system represents true differences on a substantive level. **Self-report, in general, is an excellent way to measure how an individual responds to a questionnaire. It is not necessarily a measure of the phenomenon you are seeking to learn about, unless that phenomenon is primarily about how people respond to questionnaires!** Case in point, the 2016 presidential election!

If you remember anything about social measurement, the above should be it. Never, ever assume that the behavioral action of someone responding on a questionnaire necessarily correlates to what you are ultimately seeking to measure. Even lying aside (good scales will have lie scales built in to attempt to detect dishonesty, to varying

degrees of success), human beings are not necessarily **aware of their own behavior**, which is why you are presumably studying them in the first place. People might also **project** in the opposite direction. For instance, for individuals trying to overcome their attachment to money, they may minimize how much they value money on a questionnaire. The truth is that the person likes money a great deal, but in an effort to distance themselves from the addiction, responds with more of their “**intention**” than with what is actually true in reality. Such confounds exist aplenty in **psychological measurement**. The only thing you can be sure of, for certain, is that the person responded that he or she did not value money to a great deal because that was what was circled on the page. If you are able to correlate that to lifestyle, expenditures, and other things money-related, you may be able to substantiate that measurement with behavioral observation. Self-report behavior can often be quite different from other behavior, but it is usually the “other behavior” that we consider more valid. As a scientist, your job is to figure out a way to tap into that and measure it properly (good luck, it will not be easy!).

Despite our caveats on the measurement of difficult phenomena, still, as mentioned, attempting to measure on a continuous scale rather than a dichotomous or polytomous one is probably usually the best first approach or attempt. You can also “bin” your data afterward if the results suggest that the underlying variable is more categorical than you once thought. For example, asking people how many car accidents they got into this month, while you may think it might contain a range of numbers, it is likely and more realistic that it will contain those having zero car accidents and those having one. That is, the true underlying variable that is even measurable is likely to be **binary** in this case. At the same time, some phenomena, at least at first glance, naturally implies discrete measurement. Whether you passed or failed a course in college is naturally discrete, though your grade is not necessarily so, as the measurement of “success” in the class can be broken down further into letters A, B, C, and so on, just as the letter grades can be broken down into actual measured scores (e.g. 70, 75, 80, etc.). The lumping into categories is probably not best here from a measurement perspective, and statistical analyses of such data would be best conducted on the actual measured scores. It is much easier to “bin” after the fact than initially. So, in an exploratory sense, you might first “try” for continuity, and then, if that fails, settle for dichotomous or polytomous coding.

8.2 The Generalized Linear Model

Based on our discussion thus far, it may seem at first glance that when working with a non-continuous response variable, one could never even conceive of implementing a model that comes anywhere near close to resembling the linear regression model of previous chapters. Since the regression model assumes continuity of the response, it would appear that we would require a completely different model to analyze response variables that are binary or polytomous. While there is truth to this, as we will see, the stepping stones to this new class of models will surprisingly have parallels to the original regression model. The **generalized linear model** (McCullagh & Nelder, 1990) is **a class of models that incorporates binary and polytomous models into its framework, and applies a transformation to the binary or polytomous variable**

such that it resembles a variable that is linear. Hence, the generalized linear model can be thought of as a way of “linearizing” a non-linear model such that it behaves more similarly to a linear model. In the case we will consider shortly, the **logistic regression model**, a special case of the **generalized linear model**, will effectuate a transformation on something called the **odds**, and generate a linear response from this transformation. From there, as we will see, the interpretation of the resulting coefficients will be similar to that of the traditional linear model. Hence, the name “generalized” linear model naturally encompasses the linear model as a special case, but, as we will see, also includes non-linear responses. The most popular of these special cases is the logistic model, which we now consider in some detail.

8.3 Logistic Regression for Binary Responses: A Special Subclass of the Generalized Linear Model

Binary logistic regression is one of the simpler and “easier to digest” of the generalized linear model class, and is also probably the most popular in the sciences. If you understand the basics of this model, understanding other models in the generalized linear model class should not be that difficult. Logistic regression works by transforming a binary response into a variable that is continuous and operates its analysis on this newly defined variable, making it much more similar to ordinary least-squares (OLS) regression than when the variable was binary.

The first step to understanding logistic regression is to define what is known as the **odds**:

$$\text{odds} = \left(\frac{p}{1-p} \right)$$

where p is the probability of an event occurring, while $1-p$ is the probability of an event not occurring. The odds are computed for a **mutually exclusive event**, meaning that the event occurs or does not occur on any given trial. One of these two possibilities must occur. And since by the rules of theoretical probability the probability of the total sample space must equal 1.0, it stands therefore that $p + (1-p) = 1.0$. That is, the probability of the event occurring and not occurring on a single trial must sum to unity. You can think of odds as being “centered” at a value of 1.0, which in this case implies that the probability of the event occurring is equal to the probability of the event not occurring. Notice that this holds only if p is equal to 0.5, since

$$\text{odds} = \left(\frac{p}{1-p} \right) = \left(\frac{0.5}{1-0.5} \right) = 1.0$$

Hence, if the odds of an event occurring are 1 to 1, then it implies that the probability of that event is equal to 0.5. It is easy to see that as p increases, the odds in favor of the event are greater than 1.0. For example, for $p = 0.6$, the odds in favor of the event are $0.6 / 0.4 = 1.5$ to 1. For $p = 0.9$, the odds in favor of the event are $0.9 / 0.1 = 9$ to 1. On the opposite end, as p decreases, the odds in favor of the event become less than 1.0.

The odds are useful here because our response variable is dichotomous and the categories are mutually exclusive. For instance, if we are trying to predict recovery from COVID-19 based on several predictors, we may be interested in the odds of recovering

versus not recovering, or the odds of surviving versus not surviving. In both of these examples, the response variable has two possibilities, and the occurrence of one possibility precludes the other. That is, if you recover, then you cannot “not recover.” If you survive, then this implies you did not die. This is where the concept of mutual exclusiveness fits into things here.

Now, the odds are one thing, but they are built on a binary principle of p vs. $1 - p$. Regression analysis is not ideally suited to operate on a binary response variable. However, not all hope is lost. What if we were able to somehow transform the odds into something that is linear? This is exactly the approach we will take for the logistic regression. That is, our goal will be to transform the odds into a variable that we could perform a regression on, analogous to performing a least-squares regression on a continuous variable. The key to logistic regression then will be to transform the odds into something that is linear, which we will call the **log of the odds**. More specifically, we take the **natural log of the odds**:

$$\ln(\text{odds}) = \ln\left(\frac{p}{1-p}\right)$$

The natural log of the odds, given as **ln(odds)**, is also known as the **logit**. Referring to our previous example, for an odds of 1.0, the log of the odds is therefore equal to 0. Hence, a logit of 0 represents “even odds” and a probability of 0.5 for the even occurring. Notice that as p is greater than 0.5, the log of the odds will be greater than 0. For p less than 0.5, the log of the odds will be negative.

Hence, we have seen how we can go from probabilities to odds and from odds to the logged odds, or the logit, for short. The key point to logistic regression is in this transformation. Once we obtain the logit, we can interpret it as we would an ordinary regression coefficient, only now we are no longer interpreting the coefficient in the original units of the response variable. Instead, we are interpreting it in the units of the logit. For example, in a previous chapter we regressed verbal score onto quantitative score and obtained a regression coefficient of 0.56. This coefficient was computed on the original units of the variable, and hence our interpretation was that for a one-unit increase in quantitative, we could expect, on average, a 0.56-unit increase in verbal. Had the analysis instead been one of logistic regression, the 0.56 would be in units of the logit and our interpretation would have been for a one-unit increase in quantitative, we would expect, on average, a 0.56 unit increase in the logit of verbal. This is what we mean by the coefficient being computed on different units. Now in the case of verbal on quantitative, we would not be doing a logistic regression in the first place, and hence would never be interpreting the 0.56 figure in terms of a logit increase. Our example here is merely meant to compare how OLS regression compares to logistic in interpretation.

We can easily demonstrate the transformation of probabilities to odds and odds to logits directly in Python. We start with a probability of 0.5 and compute the odds:

```
odds = 0.5 / (1 - 0.5)
odds
Out[102]: 1.0
```

Notice that, here, the odds are computed as p to $1 - p$, where p in this case is equal to 0.5, and hence $1 - p$ is equal to 0.5 also (given as `1 - 0.5` in the code). Not surprisingly,

the odds are equal to 1.0, as we previously said they would be for a probability of 0.5. We now compute the log of the odds. For this, we will use **numpy**:

```
import numpy as np
np.log(odds)
Out[106]: 0.0
```

Python tells us the log of the odds are equal to 0, which again, makes perfect sense. Notice also that if the probability of an event occurring is equal to exactly 1.0, then the odds, computationally at least, are **undefined**, and Python will let us know such by the following:

```
odds = 1.0/(1.0-1.0)
ZeroDivisionError: float division by zero
```

The odds are undefined because **division by 0 is undefined**. Note carefully that this is different from the odds equaling 0. That is, “undefined” does not equate to equaling zero, and can be easily misunderstood. If you try dividing 0 by 2 on your pocket calculator, you will get 0. On the other hand, if you divide 2 by 0, you will get an error message, “E” informing you the division cannot be executed. That is, it is undefined. On a more theoretical level, one might say that as the probability of an event nears 1.0, the odds approach **infinity**. However, since infinity is more of a **concept** than an actual number, speaking of $1/0$ as an actual number for the odds is very awkward. However, as $p \rightarrow 1$, the odds $\rightarrow \infty$.

If the probability of an event is equal to 0, then the odds are equal to zero:

```
odds = 0.0/(1.0-0.0)
odds
Out[109]: 0.0
```

If we can go from odds to logits, we should be able to go in the reverse direction as well, from logits to odds. However, how does one “undo” a logit? You may recall that the inverse operation of taking a logarithm is to **exponentiate**. For example, consider the logarithm of the number 8 to base 2:

$$\log_2(8) = 3$$

The log to base 2 of the number 8 is equal to 3. Why? Because 2 raised to the exponent 3 is equal to 8. In this case, 2 is the base and 3 is the exponent we raise to, to get 8. We can easily demonstrate this in Python:

```
2**3
Out[112]: 8
```

Notice that in Python, exponentiation is denoted by ******. This is contrary to R software, where it is denoted by **^**. Notation for mathematical operations in software is unfortunately not standardized across programs so you must always be sure of what you are computing. You may think you are computing one thing, but in reality it may be something entirely different! Be slow and cautious in your work as misunderstandings regarding what the program is actually computing are relatively easy to come by. **Always double (and triple) check.**

Above, we exponentiated to base 2 simply as an example to demonstrate or remember how to compute a logarithm. In logistic regression, we will not be exponentiating to base 2, but instead to base e , which is the **exponential function**, where e is a constant equal to approximately 2.71828 (called “Euler’s number”). We emphasize “approximately” here because e is an **irrational number**, and hence has no terminating or periodic (i.e. repeating) decimal. Hence, if you want to obtain the odds from the logit, then you should **exponentiate the logit**. Again, as a simple example, suppose the odds in favor of the event are 2 to 1. The log of the odds are therefore

```
np.log(2)
Out[113]: 0.6931471805599453
```

That is, the logit is 0.69. Now, to get the odds back, we exponentiate the logit, which means we raise e to the power of 0.69:

```
np.exp(0.6931471805599453)
Out[116]: 2
```

We can see that when we exponentiate the logit, we get the number 2, which is the original odds. Taking the log and exponentiating are reverse operations. We now have all the technical precursors in place to survey logistic regression in Python, which we do next.

8.4 Logistic Regression in Python

The data we will use comes from the accident and explosion of space shuttle **Challenger** in 1986. On January 28, 1986, the space shuttle lifted off from Cape Canaveral, Florida at 11:39 am, EST. Shortly into the flight, the shuttle exploded and all crew were lost in the tragedy. A follow-up investigation revealed that the o-rings on the shuttle’s boosters that are designed to keep the fuel from seeping out of the booster, failed, allowing fuel to ignite and cause the disaster (Figure 8.1). Presumably, the o-rings expanded due to the temperature at which the shuttle was launched, which

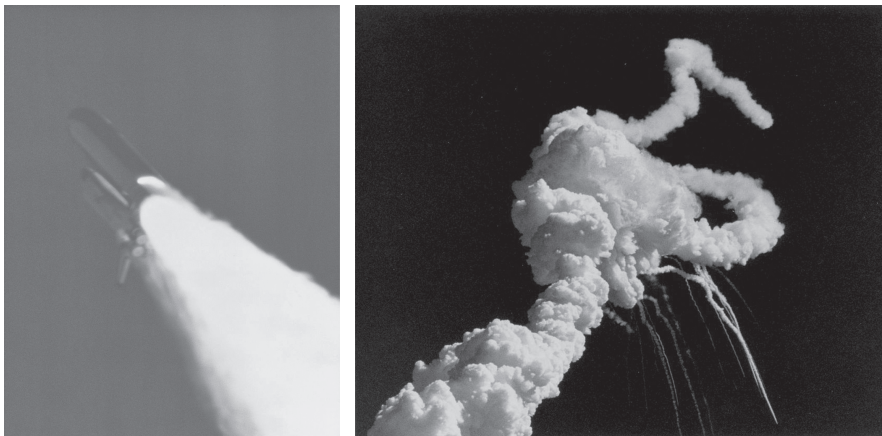


Figure 8.1 Challenger shuttle disaster of 1986. Challenger in flight with flame exiting the right booster (left). Explosion of the shuttle moments later (right).

was a frigid 31 degrees, much colder than any previous launch. The cold temperature apparently caused the o-rings to expand, allowing fuel to leak out onto the main booster, causing the explosion.

The following are the Challenger data that include data from prior launches. These are measurements as to whether or not there was any “blow-by” that was recorded as a failure. A blow-by is crudely defined as gases seeping beyond the seal of the booster. Hence, in the following, “1” is denoted as a **failure** (i.e. there is blow-by), while “0” is denoted as **no failure**. The temperature variable (`temp`) is in degrees Fahrenheit. We build the dataframe in Python (we print only the first six cases in the output):

```
import pandas as pd
data = {'oring' : [1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0,
0, 1, 0, 0, 0, 0, 0],
'temp': [53, 57, 58, 63, 66, 67, 67, 67, 68, 69, 70, 70, 70, 70,
72, 73, 75, 75, 76, 76, 78, 79, 81]}
df_challenger = pd.DataFrame(data)
print(df_challenger)

   oring  temp
0      1    53
1      1    57
2      1    58
3      1    63
4      0    66
5      0    67
```

We first confirm the number of failures (1) and no failures (0) on the entire data set (not just the six data points listed above):

```
df_challenger['oring'].value_counts()
Out[124]:
0      16
1       7
Name: oring, dtype: int64
```

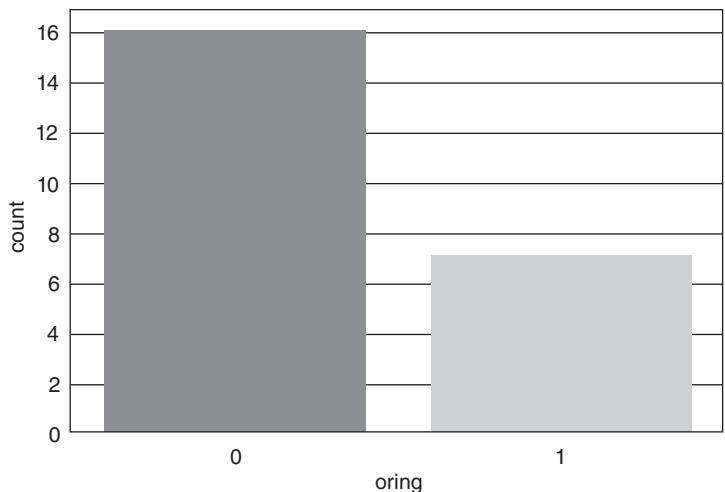
Hence, we can see from this that we have a total of 16 booster successes and 7 times the booster failed. That is, in this data set, 16 times there was no “event” of seepage or blow-by, and 7 times there was. A plot will allow us to visualize the distribution of counts. For this, we will use **seaborn**:

```
import seaborn as sns
sns.countplot(x='oring', data = df_challenger, palette='hls')
Out[126]: <matplotlib.axes._subplots.AxesSubplot at 0x1cc59710>
```

We can see the counts for o-ring failures (1) represented in this bar graph (below). We now run the logistic regression. Though there are a few ways of computing a logistic regression in Python, we will use **statsmodels** (sklearn can also be used):

```
import statsmodels.api as sm
```

We will add a constant to our model via `sm.add_constant()`, define our `y` vector as equal to the `oring` variable, and `x` as equal to `temp`:



```
y = df_challenger['oring']
X = df_challenger['temp']
X_const = sm.add_constant(X)
```

We now specify our model as follows, and request model fit statistics through `model.fit()`:

```
model = sm.Logit(y, X_const)
results = model.fit()
Optimization terminated successfully.
    Current function value: 0.441635
    Iterations 7
print(results.summary())
```

Logit Regression Results						
=====						
Dep. Variable:	oring		No. Observations:	23		
Model:	Logit		Df Residuals:	21		
Method:	MLE		Df Model:	1		
Date:	Sun, 20 Sep 2020		Pseudo R-squ.:	0.2813		
Time:	12:14:39		Log-Likelihood:	-10.158		
converged:	True		LL-Null:	-14.134		
Covariance Type:	nonrobust		LLR p-value:	0.004804		
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	15.0429	7.379	2.039	0.041	0.581	29.505
temp	-0.2322	0.108	-2.145	0.032	-0.444	-0.020
=====						

The intercept of the regression equation is equal to 15.04, while the parameter estimate for `temp` is equal to `-0.23`. The model reports a pseudo R-squared value (McFadden’s pseudo R-squared) of 0.2813, where “pseudo” here implies the statistic

should not be interpreted one-to-one with a traditional R-squared value as one would have in an OLS regression. That is, it is not truly a “variance explained” measure, but nonetheless can be used as a general measure of effect size for the model. For details on pseudo R-squared statistics, see Cohen et al. (2002).

We can see that `temp` is statistically significant ($p = 0.032$). The model equation is thus:

$$\text{predicted logit} = 15.0429 - 0.2322(\text{temp})$$

This is the estimated equation for predicting the response variable, which in this case recall is the **logit**. Recall that the response variable here is the logged odds (i.e. the logit) and not the naturally-occurring original binary variable we started out with. Let us use the equation to predict some logits for different values of `temp`, which recall is “`x`” in our model set-up. First, we will enter our equation into Python:

```
predicted_logit = 15.0429 - 0.2322*X
```

For example, our first `temp` value in our data file is 53. The predicted logit is thus:

```
predicted_logit = 15.0429 - 0.2322(X)
                = 15.0429 - 0.2322(53)
                = 2.7363
```

That is, 2.7363 is what we would expect for a `temp` of 53. The other values follow:

```
predicted_logit
Out[229]:
      temp
0    2.736648
1    1.808432
2    1.576378
3    0.416108
4   -0.280054
5   -0.512108
6   -0.512108
7   -0.512108
8   -0.744162
9   -0.976217
10  -1.208271
11  -1.208271
12  -1.208271
13  -1.208271
14  -1.672379
15  -1.904433
16  -2.368541
17  -2.368541
18  -2.600595
19  -2.600595
20  -3.064703
```

```
21 -3.296757
22 -3.760865
```

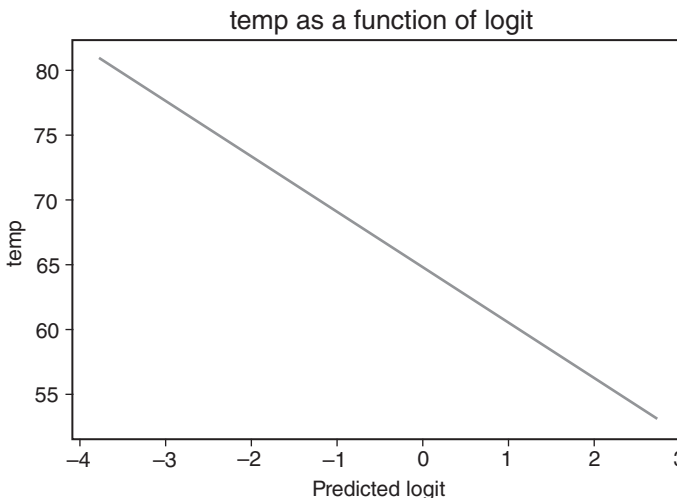
Notice that we go from positive numbers toward zero, then increasingly into negative numbers of greater absolute magnitude. The reason for this is because our temp data follows a similar trajectory (from smaller values of temp to larger). Indeed, a Spearman rho should detect this **monotonicity** and yield a correlation in absolute magnitude of 1.0:

```
from scipy import stats
stats.spearmanr(predicted_logit, X)
Out[232]: SpearmanrResult(correlation=-1.0, pvalue=0.0)
```

We see that the obtained correlation is equal to -1.0 . The p -value is irrelevant, as we simply wanted to demonstrate that as values of temp decrease, values of the logit increase (and vice versa). A plot of this relationship clearly reveals the perfect linear trend:

```
import matplotlib.pyplot as plt
import numpy as np
plt.plot(predicted_logit, X)
plt.xlabel("predicted logit")
plt.ylabel("temp")
plt.title("temp as a function of logit")

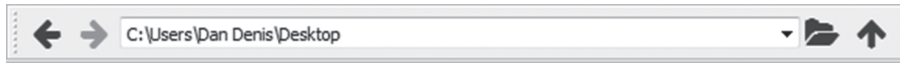
Out[38]: Text(0.5, 1.0, 'temp as a function of logit')
```



8.5 Multiple Logistic Regression

As when generalizing from simple linear regression to multiple, we can likewise generalize the simple logistic model to a multiple logistic regression. The data for this example comes from the **ISLR library** in R software, and consists of percentage

returns for the S&P 500 stock index over the course of 1,250 days in the period beginning 2001 to the end of 2005. Let us first obtain the data and then describe its variables. If the data are not loading below, make sure Spyder, or whatever environment you are using, is searching the correct location for the file; for example, the `Smarket.csv` file for the following data I saved to my Desktop; hence, the top right of the Spyder interface looks like this:



```
import pandas as pd
import numpy as np
import statsmodels.api as sm

df = pd.read_csv('Smarket.csv', index_col=0, parse_dates=True)
df.head()
Out[414]:
```

	Lag1	Lag2	Lag3	Lag4	Lag5	Volume	Today	Direction
Year								
2001-01-01	0.381	-0.192	-2.624	-1.055	5.010	1.1913	0.959	Up
2001-01-01	0.959	0.381	-0.192	-2.624	-1.055	1.2965	1.032	Up
2001-01-01	1.032	0.959	0.381	-0.192	-2.624	1.4112	-0.623	Down
2001-01-01	-0.623	1.032	0.959	0.381	-0.192	1.2760	0.614	Up
2001-01-01	0.614	-0.623	1.032	0.959	0.381	1.2057	0.213	Up

Here, `Lag1` through `Lag5` are the corresponding previous trading days. For each date, the percentage returns for each trading day are given. The volume (in billions) of the shares traded on the previous day of trading are also given. The variable `Today` is the percentage return on the given date, and `Direction` is the variable `Today`, but binary-coded into whether the market was up or down on the given day. Our goal will be to predict `Direction` based on the five lag variables as well as volume. Hence, our function statement is the following:

$$\text{Direction} = \text{Lag1} + \text{Lag2} + \text{Lag3} + \text{Lag4} + \text{Lag5} + \text{Volume}$$

We will again use `statsmodels` to perform the logistic regression:

```
import statsmodels.formula.api as smf
```

The following is the model we would like to run:

```
Direction ~ Lag1+Lag2+Lag3+Lag4+Lag5+Volume
```

To run the model, we use the `glm` function, which stands for **generalized linear model**, and specify the family as `binomial()`. To get the correct coefficients, we need to also adjust how Python is coding the binary response, and add an intercept:

```
change = np.where(df['Direction']=='Up', 1, 0)
model_constant = sm.add_constant(df[['Lag1', 'Lag2', 'Lag3',
'Lag4', 'Lag5', 'Volume']])
model = sm.GLM(change, model_constant, family=sm.families.
Binomial()).fit()
```

```
print(model.summary())
```

```
Generalized Linear Model Regression Results
```

```
=====
```

Dep. Variable:	y	No. Observations:	1250
Model:	GLM	Df Residuals:	1243
Model Family:	Binomial	Df Model:	6
Link Function:	logit	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-863.79
Date:	Mon, 21 Sep 2020	Deviance:	1727.6
Time:	01:02:59	Pearson chi2:	1.25e+03
No. Iterations:	4		
Covariance Type:	nonrobust		

```
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	-0.1260	0.241	-0.523	0.601	-0.598	0.346
Lag1	-0.0731	0.050	-1.457	0.145	-0.171	0.025
Lag2	-0.0423	0.050	-0.845	0.398	-0.140	0.056
Lag3	0.0111	0.050	0.222	0.824	-0.087	0.109
Lag4	0.0094	0.050	0.187	0.851	-0.089	0.107
Lag5	0.0103	0.050	0.208	0.835	-0.087	0.107
Volume	0.1354	0.158	0.855	0.392	-0.175	0.446

```
=====
```

We interpret the primary features of the output:

- The dependent variable is `Direction`, with categories `down` and `up`. We are modeling the category “`up`”.
- The model family (`Binomial`) and link function (`logit`) confirm that we are working with a logistic regression.
- The total number of observations in the data file is equal to 1250.

Now, since this is a multiple logistic regression having many predictor variables, it is imperative to emphasize the interpretation of parameter estimates in the context of the model. For example, the coefficient for `Lag1` of `-0.0731` is interpreted as follows:

For a one-unit increase in `Lag1`, we expect, on average, the logit on the response variable to decrease (because of the negative sign) by 0.0731 units given the context of the model, or, otherwise said, given the simultaneous inclusion of predictors `Lag2` through to `Lag5` as well as `volume`.

Recall that just as was the case in least-squares regression, it would be incorrect to omit a statement about the “context of the model.” That is, **the decrease of 0.0731 is not an expected decrease for `Lag1`**. Rather, it is an expected decrease for `Lag1` in the **context of also including `Lags 2` through `5`, as well as `volume`**. If you fail to mention the context of a model and simply cite a coefficient for a given predictor, you are misleading your audience and reporting incorrect information. **All models are context dependent. You must include the context of the model you ran when interpreting individual coefficients.**

The above are in units of the logit, or the “logged odds.” We may prefer to interpret things in terms of **odds**. To get the odds for each predictor, recall that we are to exponentiate the logit. For example, for `Lag1`, the computation is

```
import numpy as np
np.exp(-0.0731)
```

```
Out[225]: 0.9295078745906393
```

The odds are thus 0.929 for `Lag1` holding all other variables in the model constant (or again, “given the context of the model”). Since an odds of 1.0 represents even odds, the number 0.9295 represents a **decrease**. That is, for a one-unit increase in `Lag1`, the likelihood of the event decreases (i.e. it is below a probability of 0.50 for our data). This is challenging to make sense of in terms of odds. Probabilities here will be much more intuitive. Fortunately, we can also convert this odds to a probability. We do this by dividing the odds by **1 + odds**:

```
prob = np.exp(-0.0731)/(1 + np.exp(-0.0731))
```

```
prob
```

```
Out[227]: 0.4817331335265174
```

As we can see, the probability is 0.4817 of being in the group labeled 1 (i.e. “1” corresponding to an upward direction in our data). The probability of 0.4817 makes good sense too, since the odds are just below 1.0. That is, had our computed probability come out much higher for some reason (e.g., 0.90), we would know that something was amiss, since the odds are actually lower than 1.0. It is always a good idea to conduct these kinds of informal **confirmatory checks** on your computations from time to time to make sure they agree with one another. If anything is amiss, “unreasonable” numbers sometimes appear and signal that a recheck of your work is in order. If something “doesn’t seem right,” it usually isn’t (it is difficult enough ensuring things are correct even when they seem right!).

8.5.1 A Model with Only Lag1

As we emphasized earlier, it is incorrect to interpret parameter estimates without reference to the model. Why? Because if we did not include the other predictors, the parameter estimate for `Lag1` would change! Not convinced? We will include only `Lag1` to demonstrate this (we print only the model coefficients):

```
model_constant = sm.add_constant(df[['Lag1']])
model = sm.GLM(change, model_constant, family=sm.families.
Binomial()).fit()
print(model.summary())
```

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
const          0.0740         0.057        1.306      0.191      -0.037         0.185
Lag1          -0.0702         0.050       -1.404      0.160      -0.168         0.028
=====
```

Note that the coefficient for `Lag1` has changed from its original value, from an estimate of -0.0731 in the original model to -0.0702 when used as the only predictor. The change in this case is slight, but the point is that **these coefficients do habitually change**. The p -value has changed from 0.145 to 0.160 , as well as the confidence limits from the lower limit -0.171 to upper limit 0.025 in the full model to the lower limit -0.168 to the upper limit 0.028 in the one-predictor case. These changes can become quite impactful in other models. Again, whatever model you run, it should always be interpreted in the context of other predictors entered (or not entered) into the model. This is why **experimental designs** are so favored over correlational ones; we stand a better chance at controlling for these confounds. In correlational research, virtually almost anything and everything under the sun could be impacting your model. The problem of **omitted variables** in correlational designs is so relevant that many experimentalists will often not give correlational models much thought at all. There is a reason why experiments are usually considered the “gold standard” of evidence. Evidence that a COVID-19 vaccine works based on correlational evidence, for instance, is simply not good enough. Having said that, many times correlational designs are the best we can do because experimental control is either not do-able or entirely unethical. So long as their limitations are understood and appreciated, these models can still be very useful and should not be automatically disregarded.

Instead of obtaining output for the entire model as above, we can also request Python to print only information we immediately desire. For example, if you try `print(model.params)` and `print(model.pvalues)`, you will obtain the same information as earlier.

8.6 Further Directions

In this chapter we have surveyed the logistic regression model as a special case of the more generalized linear model. There are many more **link functions** that the generalized linear model can accommodate, such as a **Poisson link** or **negative binomial link**, among others. Fox (2016) is an excellent source for following up this chapter with a full treatment of the generalized linear model. A technique in machine learning and computer science, that of **support vector machines**, is yet another method whose purpose is to classify observations into categories, a kind of competitor to both logistic regression and discriminant analysis. For details, see James et al. (2013), or, for a much deeper treatment, see Izenman (2008).

Review Exercises

1. From a purely mathematical point of view, classic **logistic regression** features a binary response variable. From a scientific perspective, however, the decision of whether to consider a response variable binary or not is much more complicated. Why? Discuss how measurement issues are at the forefront of such decisions.
2. Why is it the case that true **continuity** is a **philosophical notion** and not a **scientific** one? Explain.

3. In a sentence or two, describe how **linear models** can be subsumed under the more **generalized linear model** framework. For example, how is a linear model also a generalized linear model?
4. If the **probability** of rain today is 0.80, and it must either rain or not rain, what are the **odds** of rain?
5. Justify that the odds are “**centered**” at a value of 1.0. That is, why is it that an odds of 1.0 indicates an even probability for or against an event from occurring?
6. Does the **odds** have an upper bound? Why or why not? Justify.
7. Is an **odds** less than zero possible? Justify by reference to the formula for the odds.
8. Can a **logit** be negative? Why or why not? Show, using the formula, whether this can or cannot be the case.
9. For what value of probability are the odds **undefined**? What does it mean to say they are undefined anyway?
10. Consider the **Heart Failure Clinical Records** data set in the **Machine Learning Repository**. Perform a logistic regression attempting to classify sex (woman or man) based on the age of the patient (in years) and serum sodium in the blood. Interpret the odds of heart failure given that you are a male relative to if you are a female in the context of serum sodium in the blood. What is the probability of heart failure given that you are a male compared to if you are a female?